From Person-Specific Dynamics to Population Inference: A Fast, Parallelized Meta-Analytic Structural Equation Modeling Framework for Integrating Complex Multilevel Dynamic Models

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Abstract

Intensive longitudinal research increasingly seeks population-level inference on within-person dynamics without sacrificing person-specific estimation. We introduce MetaVAR, a two-stage meta-analytic structural equation modeling (MASEM) framework for multilevel dynamic systems that extends prior idiographic two-stage work to multivariate settings. In Stage 1, person-specific vector autoregressive (VAR) models are fit to obtain each individual's parameter vector and corresponding sampling covariance matrix. In Stage 2, these parameter vectors are synthesized using a multivariate random-effects meta-analytic SEM, which propagates first-stage uncertainty and enables system-level inference on the full dynamic parameter system. MetaVAR yields population summaries of average dynamics, between-person heterogeneity, and cross-parameter covariation, and it can accommodate constraints as well as extensions involving covariates or distal outcomes. We evaluate MetaVAR in a Monte Carlo simulation against a Bayesian multilevel VAR estimator and a naïve fit-many-then-summarize approach. MetaVAR performs especially well in recovering between-person heterogeneity and in producing better-calibrated interval estimates for heterogeneity parameters, whereas one-step hierarchical Bayesian modeling using relatively uninformative priors remains competitive for population-average effects but not random effect variance-covariance parameters. We illustrate the approach using ecological momentary assessment data on negative and positive affect, showing that MetaVAR recovers interpretable average dynamics together with meaningful heterogeneity in affective set points, temporal persistence, cross-lagged coupling, and innovation structure.

Keywords: vector autoregressive model, multilevel model, random-effects meta-analysis, structural equation modeling

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Please use the **todo** tag in this format for your comments to make version control easier.

SMC: I love your intro, Jek. I do think you should think a bit more about your main goal in this paper. Do you want to introduce MetaVAR (a package, focus is primarily on VAR only), or the framework (broader use of MASEM for fitting complex multilevel dynamic models). If the latter, the current title is good. If the former, then you should have MetaVAR somewhere in your title to facilitate citations by others.

Intensive longitudinal designs, such as ecological momentary assessment (EMA), experience-sampling, daily diary, ambulatory assessment, and measurement-burst protocols, have made it increasingly feasible to collect dense data within-person in everyday contexts (Bolger & Laurenceau, 2013; Bolger et al., 2003; Csikszentmihalyi & Larson, 1987; Nesselroade, 1991; Shiffman et al., 2008; Sliwinski, 2008; Trull & Ebner-Priemer, 2013). These designs have expanded opportunities to study psychological processes as dynamic systems, in which within-person trajectories evolve over time and individuals differ meaningfully in the parameters that govern those trajectories (Asparouhov et al., 2018; Bringmann et al., 2013, 2016; Fisher et al., 2011; Hamaker et al., 2018; Rowland & Wenzel, 2020). This shift reinforces a long-standing methodological point: inferences about within-person dynamics cannot be assumed from between-person variation, and questions about regulation, coupling, and stability require models that distinguish idiographic (person-specific) from nomothetic (population-level) structure (Curran & Bauer, 2011; Fisher et al., 2018; Gates et al., 2023; Hamaker et al., 2005; Molenaar, 2004; Wang et al., 2026).

At the same time, applied researchers rarely want only a collection of person-specific estimates. They also want population-level summaries of average dynamics, between-person heterogeneity, and the ways dynamic features co-vary across people, as well as a basis for relating those person-specific dynamic signatures to covariates and distal outcomes (Beltz et al., 2016; Li et al., 2022). However, existing workflows often force a trade-off between two unsatisfactory options. One-step hierarchical models with partial pooling—including multilevel and Bayesian multilevel formulations—can support population inference, but may be computationally demanding and shrink person-specific dynamics. Fully unpooled "fit-many-then-summarize" approaches are modular and scalable, but often ignore first-stage uncertainty and the multivariate dependence among dynamic parameters, limiting valid inference on the dynamic system as a whole (Moeyaert et al., 2016; Van den Noortgate & Onghena, 2008).

This paper addresses this gap by introducing MetaVAR, a two-stage meta-analytic structural equation modeling framework (MASEM; Cheung, 2008, 2013, 2015) for multilevel dynamic models. In this manuscript, the Stage 1 idiographic model is a vector autoregressive (VAR; Hamilton, 1994; Lütkepohl, 2005) model fit separately to each individual, and Stage 2 synthesizes the resulting person-specific parameter vectors using a multivariate meta-analytic SEM with random-effects . The key methodological contribution is not simply that MetaVAR is two-stage, but that it synthesizes the full person-specific dynamic parameter vector while carrying forward the full Stage 1 sampling covariance matrix. This enables population-level inference on average dynamics, between-person heterogeneity, and cross-parameter covariation in the joint dynamic system rather than on isolated parameters one at a time.

The central question then is not only how to pool person-specific dynamic estimates but how to do so in a way that preserves inferential validity while remaining practically feasible. Existing naïve two-stage workflows lose this validity by ignoring first-stage estimation error and the multivariate dependence among person-specific dynamic parameters, whereas one-step hierarchical models can be computationally demanding and partially pool person-specific estimates toward the population mean. MetaVAR is intended for the setting in which researchers want to retain idiographic first-stage estimation while still obtaining valid multivariate population-level inference. In particular, MetaVAR is most useful when the scientific target includes not only population-average dynamics, but also between-person heterogeneity and cross-parameter covariation in the dynamic system. This practical niche is especially relevant in applications where person-specific models can be estimated independently and in parallel, making a two-stage workflow computationally attractive relative to joint hierarchical estimation. In what follows, we situate this contribution relative to existing one-step and two-stage approaches, evaluate it in a Monte Carlo simulation against a Bayesian multilevel VAR estimator and a naïve fit-many-then-summarize approach, and illustrate it using ecological momentary assessment data on negative and positive affect.

Two Dominant Strategies and a Persistent Gap

Two broad strategies dominate current practice for drawing population-level conclusions from intensive longitudinal multivariate time series. The first is hierarchical one-step (partially pooled) modeling—including multilevel and Bayesian multilevel formulations such as dynamic structural equation modeling (DSEM) and Bayesian multilevel VAR (mlVAR)—in which within-person dynamics and between-person variation are estimated simultaneously within a single hierarchical model (Asparouhov et al., 2018; Hamaker et al., 2018; Li et al., 2022). This approach is attractive in behavioral sciences because the length of the series per-person is often limited, making it useful to borrow strength across individuals while still allowing random effects in dynamic parameters (Li et al., 2022). At the same time, the estimation problem can become computationally demanding as the number of variables and random effects grows, and practical implementations often require simplifying assumptions about the random-effects structure or otherwise face feasibility and convergence constraints in higher-dimensional settings (Asparouhov et al., 2018; Epskamp et al., 2018; Li et al., 2022). Moreover, because hierarchical estimators partially pool information, person-specific dynamics are shrunk toward the population mean. Although this can improve stability and out-of-sample performance (Gelman & Hill, 2006), it also means that person-level estimates are biased toward the center and may be less suitable as direct measures of individual differences or for detecting unusual individuals (Efron & Morris, 1975).

At the other extreme are fully unpooled workflows that fit an idiographic model per person and then summarize the resulting coefficients informally, for example, by averaging. This "fit-many-then-summarize" strategy is modular and naturally parallelizable, but naïve summaries ignore that individuals' estimates differ in precision as a function of series length, missingness, and measurement quality, and simple averaging fails to weigh estimates by their uncertainty (Hedges & Olkin, 1985; Pastor & Lazowski, 2017). In intensive time-series settings, dependence and model misspecification (e.g., unmodeled trends) can also affect precision and inference if not modeled (Baek et al., 2023; Shadish et al., 2008). In multivariate dynamic systems, another limitation is that multi-stage estimation approaches do not place all parameters in a single joint model, and most approaches along this line do not retain the within-person sampling covariance matrix among dynamic parameters that is required for principled second-stage synthesis (Epskamp et al., 2018; Moeyaert et al., 2016). Valid two-stage inference therefore requires carrying forward each individual's first-stage sampling variance and, in multivariate settings, the full sampling covariance matrix rather than treating person-level estimates as error-free (Van den Noortgate & Onghena, 2008).

Hmm...this sounds similar to the first limitation. Perhaps rephrase to clarify what you mean.

These trade-offs motivate a more principled two-stage alternative: estimate person-specific dynamic models first, then synthesize those results at the population level while explicitly accounting for first-stage uncertainty and dependence among estimated parameters. The gap is therefore not simply the absence of a two-stage workflow, but the absence of a two-stage workflow that supports valid multivariate population inference on the dynamic system as a whole.

Related Two-Stage Literatures: Idiographic Meta-Analysis and Error-Corrected Pooling

MetaVAR integrates and expands existing work from two closely related bodies of literature. The first body of work is rooted in the dynamic network modeling literature.Lee and Gates (2024) advocated a two-stage idiographic meta-analytic perspective for observational time series, emphasizing the importance of estimating within-person models idiographically and then using random-effects meta-analysis to characterize population distributions of within-person effects. This perspective appropriately treats heterogeneity as a scientific target rather than a nuisance. MetaVAR adopts that same general stance, but extends the target of synthesis from selected effects to the full person-specific dynamic parameter vector. By carrying forward the full Stage 1 sampling covariance matrix, MetaVAR preserves dependence among

Too vague. Can you make this clearer. You can within-person estimates and supports inference on the joint parameter system rather than on one coefficient at a time.

Second, Zhang et al. (2025) developed an error-corrected two-stage approach for integrating individual-level dynamic factor analysis (DFA) parameters, showing that ignoring first-stage estimation error can bias second-stage random-effects estimation. MetaVAR extends this error-propagation logic by embedding~~to multivariate dynamic systems fit here as VAR models and embeds~~ the synthesis stage in an SEM framework. That formulation makes it possible to estimate average dynamics, between-person heterogeneity, and cross-parameter covariation within a single multivariate model, while also providing a foundation for model-level constraints and other extensions that we treat as future directions in this paper rather than as central demonstrations.

The move from path-specific synthesis to multivariate synthesis is not merely a technical generalization. In dynamic systems, substantive questions often concern configurations of parameters rather than isolated coefficients—for example, whether stronger autoregressive persistence co-occurs with stronger cross-lagged coupling, or whether summary features of the system vary systematically across persons. Addressing such questions requires a second stage that treats the dynamic system as the unit of inference. MetaVAR is designed for that purpose.

I would have thought that both Lee and Gates, and Zhang et al. (2025) are both one body of literature. The second body of literature I thought you would say (and think you should) is the literature on meta-SEM. Hence you are integrating work from the dynamic network (VAR variations) literature with that from the meta-SEM literature.

MetaVAR: Dynamic MASEM for System-Level Pooling With Full Error Propagation

MetaVAR is a two-stage framework that treats multilevel dynamic modeling as a meta-analytic problem in which *persons play the role of studies*. In Stage 1, each individual $i = 1, \ldots, N$ is fit with an idiographic dynamic model, yielding a vector of estimated dynamic parameters and an accompanying sampling covariance matrix. In Stage 2, these person-level parameter vectors are synthesized using a multivariate random-effects meta-analytic model expressed as a structural equation model. The key feature of this design is that the full Stage 1 sampling covariance matrix is carried forward into Stage 2. As a result, MetaVAR supports population-level inference on the *joint* dynamic parameter system, including average dynamics, between-person heterogeneity, and cross-parameter covariation.

This perspective connects directly to multivariate random-effects meta-analysis (Berkey et al., 1998; van Houwelingen et al., 1993, 2002) and MASEM (Cheung, 2008, 2013, 2015). It is motivated by questions that arise frequently in intensive longitudinal research: What is the average within-person dynamic system? How much do people differ in that system? Which dynamic features tend to co-occur across people? More broadly, the same framework provides a basis for extensions involving person-level covariates or distal outcomes, although the present manuscript focuses on the core two-stage synthesis.[1]

Stage 1 Outputs as Multivariate Effect Sizes

You need to introduce multilevel VAR model first and explain why it is interesting - like relate to actor-partner model, RICLPM, etc.. Don't keep mentioning things like AR and CR effects without letting people first know what they are.

Let $\theta_i \in \mathbb{R}^p$ denote the vector of person-i parameter estimates returned by the Stage 1 model, treated here as a multivariate set of meta-analytic effect sizes, and let $\mathbf{V}_i^2 \in \mathbb{R}^{p \times p}$ denote the corresponding Stage 1 sampling covariance matrix. The contents of $\mathbf{y}_i$ are flexible: depending on the Stage 1 model, $\mathbf{y}_i$ may include autoregressive and cross-lagged coefficients, innovation variances and covariances, set point parameters, measurement-error variances, or other person-specific features of the dynamic system. The central requirement is that Stage 1 yield $\mathbf{V}_i^2$, or a good approximation to it, so that Stage 2 can propagate first-stage uncertainty rather than treating person-level estimates as error-free.

In this manuscript, Stage 1 fits a person-specific first-order VAR model. Let $\mathbf{s}_{i,t} \in \mathbb{R}^k$ denote the observed multivariate time series for person i at occasion $t = 1, \ldots, T$. The dynamic model follows a centered VAR specification (Asparouhov & Muthén, 2018; Hamaker & Grasman, 2015),

$$\mathbf{s}_{i,t} = \boldsymbol{\mu}_i + \mathbf{A}_i \left(\mathbf{s}_{i,t-1} - \boldsymbol{\mu}_i\right) + \boldsymbol{\zeta}_{i,t}, \quad \boldsymbol{\zeta}_{i,t} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Xi}_i)\,, \tag{1}$$

where $\boldsymbol{\mu}_i$ is the person-specific set point, $\mathbf{A}_i$ collects autoregressive and cross-lagged effects governing the evolution of deviations from that set point, and $\boldsymbol{\Xi}_i$ is the innovation covariance matrix. Under stability or stationarity (e.g., all eigenvalues of $\mathbf{A}_i$ lie inside the unit circle) and mean-zero innovations, the process fluctuates around $\boldsymbol{\mu}_i$ such that $\mathbb{E}(\mathbf{s}_{i,t}) = \boldsymbol{\mu}_i$.

To connect Stage 1 to Stage 2, we stack the person-specific parameters into a single vector $\boldsymbol{\theta}_i \in \mathbb{R}^p$ and take $\mathbf{y}_i = \hat{\boldsymbol{\theta}}_i$. For the present model, a convenient choice is

In the first sentence of this sectio, unclear to me why you need $\mathbf{y}_i$. Why not just use $\boldsymbol{\theta}$ and $\hat{\boldsymbol{\theta}}$. Was it just a typo?

$$\boldsymbol{\theta}_i = \left(\boldsymbol{\mu}_i', \mathrm{vec}(\mathbf{A}_i)', \mathrm{vech}(\boldsymbol{\Xi}_i)'\right)'\,, \tag{2}$$

[1] We use the label MetaVAR because the Stage 1 model in this manuscript is a VAR fit separately to each individual. More generally, however, the two-stage framework requires only that each person contribute (a) a vector of estimated model features, $\mathbf{y}_i$, and (b) an accompanying sampling covariance matrix, $\mathbf{V}_i^2$, so that first-stage uncertainty can be propagated into the population synthesis. In that sense, MetaVAR may be viewed as a special case of a broader meta-analytic template for dynamical systems.

where $(\cdot)'$ denotes matrix transpose, $\mathrm{vec}(\cdot)$ stacks a matrix by columns, and $\mathrm{vech}(\cdot)$ stacks the lower triangle of a symmetric matrix, including the diagonal. The associated Stage 1 sampling covariance matrix,

$$\mathbf{V}_i^2 = \widehat{\mathbb{V}}\left[\hat{\boldsymbol{\theta}}_i\right],$$

can be obtained from the observed information matrix, a sandwich estimator, or a bootstrap, and is then passed to Stage 2. If some components are fixed or omitted in a given application, they are simply excluded from $\boldsymbol{\theta}_i$, and the dimension p is adjusted accordingly.

Stage 2 as Multivariate Mixed-Effects Meta-Analysis in SEM Form

A general mixed-effects meta-analytic SEM can be written as (Cheung, 2008, 2013)

$$\mathbf{y}_i = \boldsymbol{\alpha}_i + \boldsymbol{\beta}\mathbf{y}_i + \boldsymbol{\Gamma}\mathbf{x}_i + \boldsymbol{\varepsilon}_i, \quad \boldsymbol{\varepsilon}_i \sim \mathcal{N}\left(\mathbf{0}, \mathbf{V}_i^2\right), \tag{3}$$

It is very confusing to me that you call Eq 3 a meta-analytic SEM. What is in (3) is not a typical SEM model, but rather, a SEM model for the person-specific parameters, $\mathbf{y}_i$. Explain that some, have some quick examples on what you can do there. And I think it would be helpful if you write $\mathbf{y}_i$ as θ_i!

with person-specific "true" effect vectors

$$\boldsymbol{\alpha}_i = \boldsymbol{\alpha} + \boldsymbol{\upsilon}_i, \quad \boldsymbol{\upsilon}_i \sim \mathcal{N}\left(\mathbf{0}, \boldsymbol{\tau}^2\right). \tag{4}$$

Here, $\boldsymbol{\alpha}$ is the grand-mean vector of the dynamic parameter system, $\boldsymbol{\tau}^2$ is the between-person heterogeneity covariance matrix, and $\mathbf{x}_i$ collects person-level covariates. The matrices $\boldsymbol{\beta}$ and $\boldsymbol{\Gamma}$ allow researchers to specify structured relations and meta-regressions defined *on the parameter system itself*.

Assuming that $\mathbf{x}_i$ is non-stochastic and that $\mathbf{I} - \boldsymbol{\beta}$ is nonsingular, the model can be written in reduced form (Bollen, 1989) as

$$\mathbf{y}_i = \left(\mathbf{I} - \beta\right)^{-1} \left(\boldsymbol{\alpha}_i + \boldsymbol{\Gamma}\mathbf{x}_i + \boldsymbol{\varepsilon}_i\right). \tag{5}$$

Therefore, conditional on $\mathbf{x}_i$, the model-implied mean vector and covariance matrix[2] are

$$\begin{aligned} \mathbb{E}[\mathbf{y}_i \mid \mathbf{x}_i] &= (\mathbf{I} - \boldsymbol{\beta})^{-1} (\boldsymbol{\alpha} + \boldsymbol{\Gamma}\mathbf{x}_i), \\ \mathbb{V}[\mathbf{y}_i \mid \mathbf{x}_i] &= (\mathbf{I} - \boldsymbol{\beta})^{-1} \left(\boldsymbol{\tau}^2 + \mathbf{V}_i^2\right) (\mathbf{I} - \boldsymbol{\beta})^{-1'}. \end{aligned} \tag{6}$$

In the special case $\boldsymbol{\beta} = \mathbf{0}$, the model reduces to a multivariate mixed-effects meta-analysis, or to multivariate meta-regression when $\boldsymbol{\Gamma} \neq \mathbf{0}$, with

$$\mathbb{E}[\mathbf{y}_i \mid \mathbf{x}_i] = \boldsymbol{\alpha} + \boldsymbol{\Gamma}\mathbf{x}_i$$

and

$$\mathbb{V}[\mathbf{y}_i \mid \mathbf{x}_i] = \boldsymbol{\tau}^2 + \mathbf{V}_i^2.$$

When no covariates are included and $\boldsymbol{\beta} = \mathbf{0}$, the model further reduces to a multivariate random-effects meta-analysis with

$$\mathbb{E}[\mathbf{y}_i] = \boldsymbol{\alpha}, \qquad \mathbb{V}[\mathbf{y}_i] = \boldsymbol{\tau}^2 + \mathbf{V}_i^2.$$

In this way, observed dispersion in $\mathbf{y}_i$ is decomposed into (a) within-person estimation uncertainty carried by $\mathbf{V}_i^2$ and (b) genuine between-person heterogeneity carried by $\boldsymbol{\tau}^2$ (Pastor & Lazowski, 2017).

This system-level formulation is the key departure from edge-by-edge pooling. MetaVAR treats the full parameter vector as the unit of synthesis, thereby retaining the within-person sampling covariance among dynamic parameters and estimating a coherent between-person covariance structure for the dynamic system.

To describe heterogeneity, we report the conventional I^2 (Higgins & Thompson, 2002) separately for each dynamic parameter. For parameter p,

$$I_p^2 = \frac{\tau_p^2}{\tau_p^2 + \tilde{v}_p}, \tag{7}$$

[2] In SEM software that supports *definition variables*, elements of the known, person-specific sampling covariance matrix $\mathbf{V}_i^2$ can be supplied as case-level data so that the model-implied covariance is recomputed for each individual (Mehta & Neale, 2005). This capability is central in meta-analytic applications because $\mathbf{V}_i^2$ is treated as a known input, whereas common mixed-effects implementations (e.g., `lmer`) do not directly support fixing a case-specific sampling covariance matrix in the likelihood (**Declercq-Jamshidi-FernandezCastilla-2020**). In `OpenMx`, this is implemented by evaluating each individual's multivariate normal log-likelihood using that individual's model-implied mean vector and covariance matrix, which may depend on definition variables (FIML; Arbuckle, 1996; Enders, 2010; Neale et al., 2015). The `metaSEM` package leverages this capability via `OpenMx` (Cheung, 2015); the `metaDyn` implementation of MetaVAR uses the same strategy to incorporate $\mathbf{V}_i^2$ and propagate Stage 1 uncertainty through Stage 2.

where τ_p^2 is the p^{th} diagonal element of the between-person covariance matrix $\boldsymbol{\tau}^2$, and $\tilde{v}_p$ is the typical Stage 1 sampling variance for that parameter,

$$\tilde{v}_p = \frac{(N-1)\sum_{i=1}^{N} w_{i,p}}{\left(\sum_{i=1}^{N} w_{i,p}\right)^2 - \sum_{i=1}^{N} w_{i,p}^2}, \qquad w_{i,p} = \frac{1}{v_{i,p}^2}, \tag{8}$$

with $v_{i,p}^2$ denoting the p^{th} diagonal element of $\mathbf{V}_i^2$ for individual i. Thus, I_p^2 gives the proportion of total variation in the estimated person-specific value of parameter p that is attributable to true between-person heterogeneity rather than Stage 1 estimation error. Values near zero indicate that most of the observed dispersion reflects sampling uncertainty, whereas values near one indicate substantial true heterogeneity.

Model-Level Hypotheses and Constraints

Can you rewrite this paragraph to give some examples of what the SEM model for parameters can do (like explaining interrelations among parameters, linkages to outcomes, etc.). Give some examples perhaps in the context of inertia, cross-coupling, perhaps leveraging your empirical motivating example.

Because Stage 2 is an SEM defined on the full parameter vector, MetaVAR is naturally suited to *model-level* hypotheses, that is, hypotheses that jointly constrain or compare multiple dynamic parameters rather than treating each coefficient as an isolated edge. This matters because many substantive predictions about regulation are multivariate by construction: for example, whether cross-lagged effects are asymmetric ($A_{1,2} \neq A_{2,1}$), whether a subset of couplings is negligible relative to others, whether inertia parameters are comparable across constructs, or whether a theoretically motivated contrast among parameters differs from zero.

In MetaVAR, such hypotheses can be expressed through standard SEM constraint machinery applied to $\boldsymbol{\alpha}$, $\boldsymbol{\tau}^2$, $\boldsymbol{\beta}$, and $\boldsymbol{\Gamma}$. This has three advantages. First, inference remains system-aware: constraints are evaluated while retaining both the within-person sampling covariance $\mathbf{V}_i^2$ and the between-person heterogeneity structure $\boldsymbol{\tau}^2$. Second, constraints may be placed not only on individual coefficients but also on functions of the dynamic system, such as stability summaries, coupling contrasts, or other composite quantities. Third, competing theory-level models can be compared using familiar SEM tools, including likelihood-based comparisons for nested models and information criteria for non-nested models.

In the present manuscript, however, our focus is on the core two-stage synthesis: person-specific dynamic parameters are estimated idiographically and then synthesized using multivariate random-effects meta-analysis, cast as meta-analytic SEM, with full propagation of each individual's Stage 1 sampling covariance matrix. Extensions involving covariates, distal outcomes, or richer second-stage structures are possible within the same template, but we treat them as future directions rather than central demonstrations here.

Evaluation and Empirical Illustration

The remainder of the paper evaluates the operating characteristics of MetaVAR and demonstrates its applied utility. We first conduct a Monte Carlo simulation study that varies sample size and per-person series length and focuses on recovery of population-average dynamics, between-person heterogeneity, and uncertainty quantification when Stage 1 sampling covariance matrices are propagated into Stage 2. We then illustrate the end-to-end workflow using intensive longitudinal affect data. We conclude with a general discussion of the method's practical implications, limitations, and extensions.

Monte Carlo Simulation Study

We conducted a Monte Carlo simulation study to evaluate the operating characteristics of MetaVAR under a controlled bivariate affect-dynamics data-generating process (DGP) and to compare its performance with two commonly used alternatives: a Bayesian multilevel VAR (BMLVAR) approach and a naïve fit-many-then-summarize approach. The simulation examined recovery of (a) population-average dynamic parameters, (b) between-person heterogeneity in those parameters, and (c) uncertainty quantification across combinations of sample size (N) and per-person time-series length (T).

Because MetaVAR is specifically designed to propagate Stage 1 sampling covariance into a multivariate second-stage synthesis, we expected its clearest advantages to emerge for recovery of between-person heterogeneity and for interval calibration for those heterogeneity parameters. At the same time, because the DGP and the Stage 1 estimation model were matched, the simulation represents a favorable, correctly specified setting for all methods.

Data-Generating Process

Data were generated using the `SimSSMVARIVary` function from the `simStateSpace` package (v1.2.15; Pesigan et al., 2025). The data-generating model was the same centered first-order VAR model used in Stage 1 (Equation 1) for person $i = 1, \ldots, N$ and occasion $t = 1, \ldots, T$.

The person-specific set points were generated as

$$\boldsymbol{\mu}_i \sim \mathcal{N}\left(\begin{pmatrix} 2.87 \\ 2.04 \end{pmatrix}, \begin{pmatrix} 1.20 & 0.46 \\ 0.46 & 1.10 \end{pmatrix}\right), \tag{9}$$

and the person-specific transition matrices were generated by drawing

$$\operatorname{vec}(\mathbf{A}_i) \sim \mathcal{N}\left(\begin{pmatrix} 0.280 \\ -0.035 \\ -0.048 \\ 0.260 \end{pmatrix}, \begin{pmatrix} 0.017 & 0.005 & 0.001 & 0.008 \\ 0.005 & 0.008 & 0.001 & 0.006 \\ 0.001 & 0.001 & 0.001 & 0.002 \\ 0.008 & 0.006 & 0.002 & 0.026 \end{pmatrix}\right). \tag{10}$$

Here, $\operatorname{vec}(\cdot)$ denotes the vectorization operator that stacks the columns of a matrix into a single column vector. For a 2×2 matrix $\mathbf{A}_i = \left(a_{jk,i}\right)$, this implies $\operatorname{vec}(\mathbf{A}_i) = (a_{11,i},\, a_{21,i},\, a_{12,i},\, a_{22,i})'$.

Because the simulation allowed between-person heterogeneity only in the set points and transition parameters, the Stage 2 fixed-effect target can be written as

$$\boldsymbol{\alpha} = \left(\mathbb{E}[\boldsymbol{\mu}_i]', \mathbb{E}[\operatorname{vec}(\mathbf{A}_i)]'\right)'. \tag{11}$$

Likewise, the Stage 2 random-effects target is the between-person covariance matrix of $(\boldsymbol{\mu}_i', \operatorname{vec}(\mathbf{A}_i)')'$. Under the present DGP, this matrix is block diagonal because the set points and transition parameters were generated independently:

$$\boldsymbol{\tau}^2 = \begin{pmatrix} \boldsymbol{\Sigma}_\mu & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Sigma}_A \end{pmatrix}. \tag{12}$$

This notation makes explicit that the fixed-effect targets correspond to $\boldsymbol{\alpha}$ and the heterogeneity targets correspond to $\boldsymbol{\tau}^2$.

To ensure a stationary VAR process for each individual, generated transition matrices were retained only if all eigenvalues of $\mathbf{A}_i$ had moduli less than 1 (i.e., spectral radius < 1). In practice, this was implemented via rejection sampling using `simStateSpace`'s `SimBetaN` helper: candidate draws of $\operatorname{vec}(\mathbf{A}_i)$ from the target multivariate normal distribution were repeatedly proposed until the corresponding $\mathbf{A}_i$ satisfied the stability constraint. Thus, the simulation targets a setting in which person-specific dynamics are heterogeneous but dynamically stable.

The innovation covariance matrix was held constant across persons and occasions,

$$\boldsymbol{\Xi} = \begin{pmatrix} 1.30 & 0.57 \\ 0.57 & 1.56 \end{pmatrix}, \tag{13}$$

so that the simulation isolates method performance for recovering heterogeneity in the set points and lagged dynamics. Fixing $\boldsymbol{\Xi}$ also simplifies interpretation because any between-person variation in estimated innovation parameters then reflects finite-sample estimation error rather than true heterogeneity in the DGP.

The parameter values were informed by prior empirical work on affective dynamics (Bringmann et al., 2013), but were not copied exactly. In particular, we parameterized the person-specific mean (set point) $\boldsymbol{\mu}_i$ rather than the intercept vector. This is substantively convenient because $\boldsymbol{\mu}_i$ directly represents each person's equilibrium level in the centered VAR parameterization, whereas the intercept in the uncentered form is a transformed quantity. The average transition matrix implies moderate positive autoregressive persistence for both variables and small negative cross-lagged effects, representing a stable bivariate regulatory system with weak reciprocal inhibitory coupling on average.

Design Factors

We primarily used a 3×3 factorial design crossing three sample sizes and three numbers of measurement occasions per individual:

$$N \in \{50, 100, 200\}, \qquad T \in \{50, 100, 200\}. \tag{14}$$

This yielded nine balanced conditions (Cases 1–9), where all individuals in a given replication shared the same series length T.

To induce heterogeneity in first-stage precision—and thus variability in the person-specific sampling covariance matrices used in Stage 2—we also included three unbalanced-T conditions (Cases 10–12). In these conditions, the sample size was fixed at $N \in \{50, 100, 200\}$ for Cases 10, 11, and 12, respectively, but the number of measurement occasions varied across individuals within the same replication. Specifically, for each individual i, the realized series length was $T_i \in \{50, 100, 200\}$, so that some individuals contributed shorter series and others longer series, producing heterogeneous $\mathbf{V}_i^2$ across persons. Analyses used each individual's realized T_i.

For each condition, we generated $R = 1{,}000$ Monte Carlo replications. In each replication, person-specific time series were generated under the DGP above, analyzed by each competing method, and summarized using the performance metrics described below.

Methods Compared

Mention some of these alternative approaches a bit more heavily in the Intro. Bayesian multilevel VAR; BMLVAR. Note that one of the contributions of this paper is to compare these to MetaVAR.

We compared three approaches representing the main inferential strategies discussed in the Introduction: (a) a two-stage method with explicit propagation of first-stage uncertainty (MetaVAR), (b) a one-step hierarchical estimator (Bayesian multilevel VAR; BMLVAR), and (c) a two-stage fit-many-then-summarize baseline that does not propagate first-stage uncertainty (hereafter, the naïve approach). Across all methods, the inferential targets were the same: population-average dynamic parameters (fixed effects) and between-person heterogeneity in those parameters (random effects), as defined by the simulation DGP.

To facilitate comparability, all methods were fit to the same simulated datasets under each (N, T) condition, and summaries were mapped to a common parameterization whenever possible (i.e., set points, VAR transition parameters, and between-person variability in those parameters). The comparison is therefore intended as a practical comparison of estimation strategies under a common target, not as a referendum on Bayesian versus frequentist inference.

MetaVAR

MetaVAR was implemented as a two-stage procedure using `fitVARMxID` (v1.0.2) for Stage 1 and `metaDyn` (v1.0.0) for Stage 2; both packages relied on `OpenMx` (v2.22.10; Hunter, 2017; Neale et al., 2015). In Stage 1, a person-specific VAR(1) model was fit separately for each individual to obtain (a) a vector of estimated dynamic parameters and (b) the corresponding estimated sampling covariance matrix. These outputs were then passed to Stage 2, where the person-level parameter vectors were synthesized using a multivariate random-effects meta-analytic model implemented in SEM form. Stage 2 treats the person-specific sampling covariance matrices as known inputs, thereby propagating first-stage estimation uncertainty into the population-level likelihood.

A key inferential feature of MetaVAR is that it operates on the full parameter vector for each person rather than pooling parameters one at a time. Consequently, MetaVAR carries forward the within-person sampling covariance among estimated dynamic parameters into Stage 2 and estimates a coherent between-person random-effects covariance structure across the parameter system. In the simulation, we evaluated MetaVAR using model-based normal-theory confidence intervals at Stage 2.

BMLVAR

Please emphasize more clearly that here and elsewhere that this is with the default uninformative priors. In the Intro, you need to say more about some known inadequacies/difficulties of selecting priors for BMLVAR, such as inadequacies of IW priors for covariance matrices, etc.

BMLVAR was fit in `Mplus` (v9; Muthén & Muthén, 2017) using DSEM with two MCMC chains and 40,000 fixed iterations. We treated this analysis as a practical benchmark under a common applied `Mplus` implementation rather than as a definitive evaluation of all Bayesian multilevel VAR specifications. Following `Mplus` guidance, Bayesian estimation was monitored using `TECH8` potential scale reduction (PSR), posterior effective sample size (ESS), and representative trace plots. In a representative simulation replication reported in the supplement, the maximum PSR reached 1.000 by the end of the run, and the 10 lowest ESS values ranged from 2,009 to 11,605. The lowest ESS values were concentrated in between-person random-effects variances and covariances for the lagged effects, indicating that posterior mixing was more demanding for heterogeneity parameters than for the fixed effects.

Because posterior credible intervals and frequentist confidence intervals are not identical in interpretation, coverage comparisons should be understood as operating-characteristic summaries under the chosen priors and MCMC settings.

Naïve

Can you not call this the naive approach? Cite Sacha Epskamp's work explicitly. Call it unweighted or something. Say more in the literature among other approaches that you compared.

As a baseline two-stage approach, person-specific VAR(1) models were fit separately and then summarized across individuals without propagating first-stage estimation uncertainty in the second stage. Specifically, the second-stage analysis treated the Stage 1 point estimates as error-free and fit only the mean vector and covariance structure of the person-specific estimates using `OpenMx` (v2.22.10; Neale et al., 2015).

This approach reflects a common practical workflow in idiographic time-series research: fit models person-by-person, extract parameters of substantive interest, and then summarize those parameters descriptively across individuals. Its advantages are modularity, transparency, and computational simplicity. However, because the second-stage summaries do not account for the within-person sampling covariance of the Stage 1 estimates, the estimated between-person covariance of the Stage 1 estimates reflects both true heterogeneity and first-stage sampling variability. Consequently, random-effects variances and covariances can be inflated, particularly when per-person series are short or Stage 1 estimation is unstable.

Comparative expectations. Relative to the naïve approach, MetaVAR was expected to improve population-level inference by carrying forward person-specific sampling uncertainty into the second stage. Relative to BMLVAR, MetaVAR trades one-step joint estimation for a modular two-stage workflow that preserves idiographic estimation while supporting multivariate population synthesis. The simulation therefore evaluates not only point-estimate recovery, but also the inferential consequences of these distinct strategies.

Performance Metrics

We evaluated performance separately for (a) population-average (fixed-effect) parameters and (b) between-person heterogeneity (random-effect) parameters.

Point-Estimate Performance

For each target parameter θ, we computed Monte Carlo relative bias and root mean square error (RMSE) across replications:

$$\begin{aligned} \text{Relative Bias}(\hat{\theta}) &= \frac{\bar{\hat{\theta}} - \theta}{\theta}, \\ \text{RMSE}(\hat{\theta}) &= \sqrt{R^{-1} \sum_{r=1}^{R} \left(\hat{\theta}_r - \theta\right)^2}, \\ \bar{\hat{\theta}} &= R^{-1} \sum_{r=1}^{R} \hat{\theta}_r. \end{aligned} \tag{15}$$

When summarizing the magnitude of relative bias without regard to direction, we used the absolute value of the relative bias: $\left|\text{Relative Bias}(\hat{\theta})\right|$.

Interval Performance (Empirical Coverage)

We evaluated uncertainty quantification using empirical coverage of nominal 95% interval estimates. For a given target parameter θ, empirical coverage was defined as

$$\widehat{\text{Coverage}}(\theta) = R^{-1} \sum_{r=1}^{R} \mathbb{I}(\theta \in \mathcal{I}_{r,.95}), \tag{16}$$

where $\mathcal{I}_{r,.95}$ denotes the nominal 95% interval estimate obtained in replication r.

For MetaVAR and the naïve approach, $\mathcal{I}_{r,.95}$ was the nominal 95% confidence interval. For BMLVAR, it was the nominal 95% posterior credible interval. Although these interval types are not identical in interpretation, the empirical inclusion rate of nominal 95% intervals provides a practically useful basis for comparing uncertainty quantification across methods.

Coverage values near .95 indicate well-calibrated interval estimation. As a secondary benchmark, Bradley's liberal robustness criterion implies that, for nominal 95% intervals, empirical coverage between approximately .92 and .97 may be regarded as acceptable (Bradley, 1978). We therefore refer to this metric generically as empirical coverage of nominal 95% intervals.

Hypothesis-Testing Performance (Empirical Power)

Because the present DGP did not include exact-zero target parameters, we evaluated inferential sensitivity using empirical power for parameters with nonzero true values rather than empirical Type I error. For a given target parameter θ and nominal level α,

$$\widehat{\text{Power}}(\theta) = R^{-1} \sum_{r=1}^{R} \mathbb{I}(\text{reject } H_0 : \theta = 0 \text{ in replication } r). \tag{17}$$

For MetaVAR and the naïve approach, rejection was based on whether the nominal 95% confidence interval excluded zero. For BMLVAR, we report the empirical detection rate based on whether the nominal 95% posterior credible interval excluded zero. Because several true effects in the DGP were small in magnitude—especially the cross-lagged effects and some heterogeneity components—empirical power is interpreted jointly with bias, RMSE, and coverage.

Simulation Targets

I don't think this subsection adds anything. Your simulation study section is getting pretty long and you still haven't gotten to the results here.

The simulation primarily focused on recovery of (a) the population-average set points and transition parameters, (b) the between-person heterogeneity in these parameters, and (c) the quality of uncertainty quantification under each method, as indexed by empirical coverage of nominal 95% intervals and empirical power.

This design tests MetaVAR's core claim: propagating person-specific Stage 1 sampling covariance into a multivariate second stage improves population-level inference relative to naïve aggregation while preserving the modularity of a two-stage workflow. More broadly, it clarifies when such a two-stage approach approximates or departs from one-step hierarchical estimation, especially for heterogeneity recovery and interval calibration.

Monte Carlo Simulation Results

Simulation performance was evaluated using relative bias, RMSE, empirical coverage of nominal 95% intervals, and empirical power. The clearest summary is that the methods differed depending on the inferential target. When results were aggregated across all parameters and design cells, MetaVAR provided the best overall balance of point-estimate performance and interval calibration. Specifically, MetaVAR yielded the smallest mean absolute value of the relative bias (|rel_bias| = 0.186), the lowest mean RMSE (0.043), and the mean coverage closest to nominal (0.926). BMLVAR was intermediate (|rel_bias| = 1.015, RMSE = 0.047, coverage = 0.760), whereas the naïve method produced the highest mean power (0.810) but also the largest mean absolute value of the relative bias (1.540), a higher mean RMSE (0.046), and the poorest coverage (0.636). Thus, when all targets are considered jointly, MetaVAR provided the strongest overall trade-off between accuracy and interval performance. However, this overall summary masked an important distinction between population-average parameters and heterogeneity parameters.

Population-Average Parameters

For population-average parameters, all three methods showed small average bias, and BMLVAR performed best overall in interval calibration. Across fixed effects, mean absolute value of the relative bias was small for all methods (0.005 for BMLVAR, 0.024 for MetaVAR, and 0.039 for naïve), with BMLVAR yielding the best mean coverage (0.962), followed by MetaVAR (0.911) and naïve (0.860). Mean power for fixed effects was high for all three methods (approximately 0.92). Thus, for recovery of population-average dynamics alone, BMLVAR was highly competitive and, in terms of fixed-effect coverage, performed best in the present simulation.

Heterogeneity Parameters

The clearest advantage of MetaVAR emerged for the heterogeneity parameters. Across random-effects variances and covariances, MetaVAR yielded substantially smaller mean absolute value of the relative bias (0.260) and lower mean RMSE (0.039) than either BMLVAR (1.482 and 0.047, respectively) or naïve (2.233 and 0.044, respectively). MetaVAR also showed markedly better interval calibration for heterogeneity parameters, with mean coverage of 0.932, compared with 0.666 for BMLVAR and 0.533 for naïve. This pattern is central to the rationale for MetaVAR: once the scientific target includes between-person heterogeneity in the dynamic system, carrying forward Stage 1 sampling covariance materially improves second-stage inference relative to treating person-specific estimates as error-free.

For compact display in Figures 1 and 2 as well as the figures in the supplement, parameter labels are abbreviated as follows: Sp = set point, AR = autoregressive effect, CL = cross-lagged effect, Var = variance, and Cov = covariance; subscripts follow the model notation, with 1 and 2 denoting the two variables. Figure 1 provides an overall summary of simulation performance by inferential target, separating population-average parameters (fixed effects) from between-person heterogeneity parameters (random effects).

Again, I am certain that this advantage has to do with the relatively uninformative priors you used from Mplus. So I worry about your using a sweeping acronym like BMLVAR. It is okay to use the acronym, but you have to clearly justify why you go with the uninformative priors when there is quite a bit of work showing that this wouldn't work (cite e.g., Li et al. for more informative alternative priors that you can implement even in Mplus). @articleLi22b, author = Yanling Li and Julie Wood and Linying Ji and Sy-Miin Chow and Zita Oravecz, title = Fitting Multilevel Vector Autoregressive Models in Stan, JAGS, and Mplus, journal = Structural Equation Modeling: A Multidisciplinary Journal, volume = 29, number = 3, pages = 452-475, year = 2022, publisher = Routledge, doi = 10.1080/10705511.2021.1911657
Papers on alternative priors for covariance matrix. See e.g.
https://www.tandfonline.com/doi/full/10.1080/00273171.2015.1065398

Figure 2 then shows parameter-level empirical coverage of nominal 95% intervals across methods and design conditions. The remaining parameter-level heatmaps for absolute value of the relative bias, RMSE, and power are provided in the online supplement. The overview in Figure 1 makes clear that the comparative pattern depended on the inferential target. For population-average parameters, all three methods performed similarly in point estimation, and BMLVAR showed the best overall interval calibration. For between-person heterogeneity parameters, however, MetaVAR showed the strongest overall performance, with lower absolute value of the relative bias and RMSE and markedly better coverage than either BMLVAR or the naïve approach. Figure 2 further shows that the clearest differences were concentrated in interval calibration: undercoverage was most severe for the naïve approach, BMLVAR was generally intermediate for heterogeneity parameters, and MetaVAR remained closest to nominal coverage across most random-effects targets and design cells.

Sample size primarily affected RMSE and power. For MetaVAR, mean RMSE decreased from 0.057 at $N = 50$ to 0.030 at $N = 200$, while mean power increased from 0.618 to 0.809. BMLVAR showed a similar pattern, with mean RMSE decreasing from 0.066 to 0.031 and mean power increasing from 0.678 to 0.842 as sample size increased from $N = 50$ to $N = 200$. The naïve method also showed lower RMSE and higher power with larger N (RMSE: 0.061 to 0.034; power: 0.738 to 0.875), but its coverage deteriorated as sample size increased (0.741 at $N = 50$ vs. 0.529 at $N = 200$). For MetaVAR, average coverage remained relatively close to nominal across sample sizes (0.933, 0.929, and 0.914 for $N = 50$, 100, and 200, respectively), whereas BMLVAR remained consistently below nominal across all N conditions (0.783, 0.756, and 0.740).

The number of measurement occasions had an even clearer impact on performance, especially for MetaVAR. For MetaVAR, mean absolute value of the relative bias decreased from 0.475 at $T = 50$ to 0.040 at $T = 200$, while mean power increased from 0.651 to 0.782 and mean coverage improved from 0.903 to 0.935. The unbalanced-T condition performed similarly to the balanced $T = 100$ condition, with mean absolute value of the relative bias of 0.099, RMSE of 0.042, coverage of 0.933, and power of 0.717. BMLVAR also improved as the number of occasions increased, although more modestly: mean absolute value of the relative bias decreased from 1.207 at $T = 50$ to 0.894 at $T = 200$, RMSE decreased from 0.049 to 0.045, and power increased from 0.698 to 0.819, but coverage remained low and essentially unchanged (between 0.754 and 0.770). The naïve method showed substantial improvement as T increased, with mean absolute value of the relative bias decreasing from 2.758 at $T = 50$ to 0.601 at $T = 200$ and coverage increasing from 0.518 to 0.764, but even in the most favorable $T = 200$ condition its average coverage remained well below nominal.

This distinction between fixed effects and heterogeneity parameters was especially pronounced under short series. For MetaVAR, average fixed-effect coverage was clearly below nominal when $T = 50$, whereas heterogeneity-parameter coverage remained much closer to the acceptable range. By contrast, the naïve method showed severe undercoverage for heterogeneity parameters throughout, especially when T was short, consistent with the expectation that ignoring Stage 1 estimation error inflates apparent between-person variability. BMLVAR, although strong for fixed effects, also showed persistent undercoverage for heterogeneity parameters, suggesting that one-step hierarchical estimation did not fully resolve recovery of the population covariance structure under the present design. This fixed-versus-random contrast is also visually evident in Figure 1, which summarizes the same pattern across all design cells.

Too short. Group it elsewhere?

Because several heterogeneity parameters were close to zero, relative bias was naturally amplified for some components. For that reason, the RMSE and coverage results are especially important for interpreting performance on those parameters.

Computational speed? Important to mention!

Taken together, these results show that MetaVAR's main advantage lies not in maximizing empirical power, but in recovering between-person heterogeneity with better-calibrated interval estimates. BMLVAR remained highly competitive for population-average dynamics, especially fixed-effect coverage, whereas the naïve approach often achieved higher apparent power at the cost of substantial bias and undercoverage. MetaVAR is therefore most attractive when the scientific target includes heterogeneity in dynamic parameters and principled uncertainty quantification is essential. These results indicate that MetaVAR's strongest advantage in the present design lies less in uniformly improving point estimation for all targets than in improving uncertainty quantification for between-person heterogeneity parameters.

Monte Carlo Simulation Discussion

Whoa !!! Your summary is too long! Just saying the same thing. Drop completely, or condense to no more than 2 paragraphs!

The simulation supports the central rationale for MetaVAR as a two-stage framework for system-level inference that propagates Stage 1 uncertainty. Its clearest advantage was in recovering between-person heterogeneity in the dynamic parameter system. Across random-effects variances and covariances, MetaVAR generally outperformed both BMLVAR and the naïve two-stage approach in bias, RMSE, and coverage, indicating that carrying the full person-specific sampling covariance matrix into Stage 2 improves inference for heterogeneity parameters.

This advantage was not universal. For population-average parameters, BMLVAR was highly competitive and in the present design provided the best overall fixed-effect coverage. The results therefore do not suggest that MetaVAR uniformly dominates one-step hierarchical estimation. Rather, they indicate that MetaVAR is especially useful when the scientific target includes the between-person covariance structure of the dynamic system, not only its average dynamics.

The naïve comparator clarifies the cost of ignoring first-stage uncertainty in two-stage workflows. Although it often showed relatively high empirical power, that apparent sensitivity came with substantial bias and interval undercoverage for heterogeneity parameters. This pattern is consistent with the expectation that treating person-specific estimates as error-free causes second-stage dispersion to absorb both true heterogeneity and first-stage estimation noise.

The design factors also clarify when MetaVAR is most useful. Increasing the number of measurement occasions improved all methods, but the gains were especially important for MetaVAR's recovery of heterogeneity parameters. This is substantively sensible: longer person-specific series improve first-stage precision, which yields more informative sampling covariance matrices and, in turn, more reliable Stage 2 synthesis. Short series remained challenging for all methods, particularly when cross-lagged effects and heterogeneity components were small.

Taken together, these findings position MetaVAR as a principled two-stage alternative to uncorrected fit-many-then-summarize workflows rather than as a universal replacement for one-step hierarchical estimation. It preserves idiographic estimation, accommodates modular first-stage workflows, and supports multivariate population synthesis with full error propagation. In the present study, that combination was most beneficial when the inferential target was heterogeneity in dynamic parameters across persons.

Several limitations qualify these conclusions. The DGP was bivariate, the innovation covariance was held constant across persons, and data were generated from the same model class used for estimation. Accordingly, the present results speak to performance under a favorable, correctly specified setting. Future work should examine higher-dimensional systems, heterogeneity in innovation processes, model misspecification, missing data, irregular measurement spacing, and Stage 1 models beyond observed-variable VARs.

A further limitation concerns the comparison with BMLVAR. Its weaker recovery of random-effects parameters may partly reflect a broader difficulty of hierarchical Bayesian estimation when variance components are small or near zero, because inference in such settings can be sensitive to prior specification and MCMC performance can degrade substantially (Gelman, 2006; Gelman et al., 2008; Papaspiliopoulos et al., 2007). This concern is especially relevant under default Bayesian settings, because work in Bayesian SEM and Mplus has shown that default priors can materially affect estimation and that alternative weakly informative priors can improve convergence when between-level variance-covariance structures are close to singular (Asparouhov & Muthén, 2024; van Erp et al., 2018). Although convergence diagnostics for a representative replication did not suggest gross nonconvergence, the lowest ESS values were concentrated in lagged-effect random-effects variances and covariances, suggesting that the most demanding part of the

Bayesian estimation coincided with the heterogeneity parameters that were most difficult to recover. The BMLVAR results should therefore be interpreted as reflecting this specific implementation rather than as a definitive statement about Bayesian multilevel VAR performance under alternative priors, additional chains, or longer runs.

Empirical Illustration: Meta-Analysis of Affect Dynamics

We illustrate the proposed MetaVAR workflow using intensive longitudinal data from the Affective Dynamics and Individual Differences study (ADID; Emotions & Dynamic Systems Laboratory, 2010), which was designed to examine how emotion dynamics unfold in everyday life and how these dynamics differ across individuals. The ADID EMA protocol sampled participants five times per day for approximately 30 days (four daytime prompts plus an evening report), yielding up to 150 measurement occasions per participant in the raw data. ADID has contributed to multiple substantive and methodological studies, including work on regime-switching dynamics, measurement/trait–state structure, network and VAR-based modeling, outlier-robust state-space modeling, and missing-data methods (**You-Hunter-Chen-etal-2019**; Chow & Zhang, 2013; Gates et al., 2023; Guo et al., 2012; Hutton & Chow, 2014; Li et al., 2022, 2024; Oh et al., 2025; Ou et al., 2023; Park et al., 2020).

This setting closely matches the goal of MetaVAR: estimate person-specific dynamic signatures in a first stage and then synthesize them at the population level in a second stage while propagating each individual's Stage 1 sampling uncertainty. In the present illustration, we focus on within-person dynamics between negative affect (NA) and positive affect (PA). Specifically, we meta-analyze the full person-specific parameter system consisting of (a) affective set points, (b) temporal dynamics encoded by the VAR(1) transition matrix—autoregressive (AR) inertia and cross-lagged coupling—and (c) the innovation covariance matrix, which captures within-occasion co-fluctuation of unexpected changes after accounting for lagged dynamics.

Substantively, NA and PA are useful targets because emotion-dynamics theories treat affect as a self-regulating system characterized by a typical level (often described as a set point), temporal persistence from one occasion to the next (often described as inertia), and possible cross-lagged coupling between affect dimensions that may reflect carryover or regulatory spillover (Houben et al., 2015; Kuppens, Allen, & Sheeber, 2010; Kuppens, Oravecz, & Tuerlinckx, 2010). MetaVAR treats heterogeneity in these features—including heterogeneity in the innovation covariance structure—as a scientific target, providing population-average inference while quantifying between-person variation and co-variation across the dynamic system.

Measures

Momentary affect in ADID was assessed using adjective ratings drawn from established affect instruments, including items from the Positive Affect and Negative Affect Schedule (PANAS; Watson et al., 1988) and circumplex-content items used to broaden coverage of activation levels (Russell, 1980). Participants rated how often they had experienced each affect descriptor since the previous assessment on an ordinal Likert-type scale, as implemented in the ADID daily diary surveys.

Cite other papers that did the same rounding to equally spaced bins. You may want to make this paragraph shorter - just cite that you are using the same exact data set and data pre-processing procedures as published elsewhere.

Using the recorded date–time stamps, we rounded responses to the nearest hour and then constructed an hourly, regularly spaced time series by inserting missing values for hours with no report. This regularization step was used to align the data with the discrete-time VAR framework employed here, so that the lagged coefficients can be interpreted as approximately hourly dependencies rather than dependencies defined on irregular intervals. Following prior ADID applications, we constructed per-occasion NA and PA composites by averaging the available negative-affect and positive-affect items, respectively, thereby treating the ordinal item ratings as approximately continuous summary measures at the composite level. To reduce the influence of slow drift across the month-long protocol and to better align the series with a stationary VAR(1) approximation, we applied within-person linear detrending to each affect composite prior to model fitting. Under this specification, person-specific set points represent typical detrended affect levels, the transition matrix encodes short-term persistence and cross-lagged coupling at the hourly timescale, and the innovation covariance matrix characterizes same-occasion co-fluctuation in unexpected deviations after accounting for lagged dynamics. These preprocessing choices therefore make the empirical target a short-term, detrended, approximately hourly affective process rather than the full raw month-long trajectory.

Results

Stage 1 estimated person-specific NA–PA dynamics for each participant, and Stage 2 synthesized these individual dynamic signatures to obtain (a) population-average dynamics, (b) between-person heterogeneity in those dynamics, and (c) systematic co-variation among elements of the parameter system.

Stage 1: Person-Specific VAR

We fit the person-specific VAR(1) model in Equation 1 separately for each participant using the `fitVARMxID` package (v1.0.2), estimating all dynamic parameters as person-specific: set points, autoregressive coefficients, cross-lagged coefficients, and innovation covariance parameters. In the processed dataset, person-specific VAR models were fit for 217 individuals, and the Stage 1 estimation converged for all individuals in the empirical analysis.

Stage 2: Multivariate Random-Effects Meta-Analysis of the Dynamic Parameter System

Stage 2 synthesized the person-specific VAR(1) parameter systems using a multivariate random-effects meta-analysis with the `metaDyn` package (v1.0.1). The meta-analytic mean vector $\boldsymbol{\alpha}$ contained nine elements, ordered as the two set points ($\mu_{\text{NA}}, \mu_{\text{PA}}$), the four VAR transition coefficients ($a_{11}, a_{21}, a_{12}, a_{22}$), and the three unique elements of the innovation covariance matrix $\boldsymbol{\Xi}$ for NA and PA ($\xi_{11}, \xi_{21}, \xi_{22}$).

Population-average dynamics ($\boldsymbol{\alpha}$). The population-average set points indicated higher typical PA than NA: $\mu_{\text{NA}} = 1.4402$ (SE = 0.0253), 95% CI [1.3906, 1.4897], and $\mu_{\text{PA}} = 2.5817$ (SE = 0.0322), 95% CI [2.5187, 2.6448]. The average temporal dynamics showed positive inertia in both affect dimensions, with $a_{11} = 0.3571$ (SE = 0.0311), 95% CI [0.2961, 0.4181] for NA and $a_{22} = 0.3167$ (SE = 0.0313), 95% CI [0.2552, 0.3781] for PA.

Both cross-lagged effects were negative on average. The NA→PA effect was $a_{21} = -0.0728$ (SE = 0.0367), 95% CI [-0.1447, -0.0009], and the PA→NA effect was $a_{12} = -0.0593$ (SE = 0.0187), 95% CI [-0.0959, -0.0227]. Thus, higher NA at one occasion predicted lower subsequent PA on average, and higher PA predicted lower subsequent NA on average. Interpreted descriptively, these estimates are consistent with reciprocal inhibitory coupling in the short-term affective system at the hourly timescale used here.

Stage 2 also estimated the population-average innovation covariance matrix in the original metric, with $\xi_{11} = 0.0333$ (SE = 0.0019), 95% CI [0.0296, 0.0370], $\xi_{21} = -0.0073$ (SE = 0.0010), 95% CI [-0.0093, -0.0054], and $\xi_{22} = 0.0609$ (SE = 0.0029), 95% CI [0.0552, 0.0666]. Thus, after accounting for lagged dynamics, the innovation variance was larger for PA than for NA, and the innovation covariance was negative, indicating that within-occasion shocks to NA and PA tended to move in opposite directions on average.

Between-person heterogeneity and co-variation ($\boldsymbol{\tau}^2$). The estimated random-effects covariance matrix $\boldsymbol{\tau}^2$ indicated substantial heterogeneity across the full parameter system. The diagonal elements (between-person variances) were 0.1373 for μ_{NA} (SD $\approx$ 0.37), 0.2227 for μ_{PA} (SD $\approx$ 0.47), 0.1867 for a_{11} (SD $\approx$ 0.43), 0.2295 for a_{21} (SD $\approx$ 0.48), 0.0599 for a_{12} (SD $\approx$ 0.24), 0.1902 for a_{22} (SD $\approx$ 0.44), and, for the innovation parameters, 0.0007 for ξ_{11} (SD $\approx$ 0.027), 0.0001 for ξ_{21} (SD $\approx$ 0.010), and 0.0016 for ξ_{22} (SD $\approx$ 0.040).

Several, though not all, off-diagonal elements were statistically distinguishable from zero, indicating systematic co-variation among components of the dynamic signature across individuals. For example, individuals with higher NA set points tended to show greater NA inertia (cov = 0.0273,

$p = .0212$). NA inertia a_{11} covaried negatively with the NA→PA coupling a_{21} (cov $= -0.0684$, $p < .001$), and PA inertia a_{22} covaried negatively with the PA→NA coupling a_{12} (cov $= -0.0224$, $p = .0130$). There was also evidence of co-variation between set points and innovation parameters, including μ_{NA} with ξ_{11} (cov $= 0.0040$, $p < .001$) and ξ_{22} (cov $= -0.0022$, $p = .0373$), and μ_{PA} with ξ_{11} (cov $= -0.0022$, $p = .0126$) and ξ_{22} (cov $= -0.0055$, $p < .001$). In addition, several covariances among the innovation parameters were statistically significant, including ξ_{11} with ξ_{21} (cov $= -0.0002$, $p < .001$), ξ_{11} with ξ_{22} (cov $= 0.0004$, $p < .001$), and ξ_{21} with ξ_{22} (cov $= -0.0002$, $p < .001$), consistent with meaningful between-person differences in the magnitude and co-fluctuation of within-occasion shocks.

Heterogeneity index (I^2). The I^2 indices were high for all nine elements of $\boldsymbol{\alpha}$, indicating that the observed dispersion in Stage 1 parameter estimates was dominated by true between-person heterogeneity rather than by within-person sampling variability. Under Equation 7, this means that the estimated between-person heterogeneity dominated the typical Stage 1 sampling variance for each parameter. Specifically, I^2 was 0.9990 for μ_{NA}, 0.9975 for μ_{PA}, 0.9636 for a_{11}, 0.9448 for a_{21}, 0.9555 for a_{12}, 0.9514 for a_{22}, 0.9994 for ξ_{11}, 0.9944 for ξ_{21}, and 0.9954 for ξ_{22}. Taken together, the $\boldsymbol{\tau}^2$ and I^2 results support the motivating premise of MetaVAR: affective set points, lagged dynamics, and innovation structure all vary meaningfully across individuals, making multivariate synthesis of the full parameter system scientifically informative.

Empirical Example Discussion

This illustration shows how MetaVAR can be used to estimate idiographic affect-dynamics parameters from intensive longitudinal data and then synthesize those person-specific signatures into population-level inferences while quantifying heterogeneity (Emotions & Dynamic Systems Laboratory, 2010; Fisher et al., 2017). In ADID, the approach yielded interpretable population-average dynamics, strong evidence of between-person heterogeneity in both set points and lagged dependencies, and substantively meaningful co-variation among elements of the parameter system (Houben et al., 2015). In addition to the familiar temporal dependencies encoded in the VAR transition matrix, MetaVAR also supports population inference on the innovation structure, which captures within-occasion co-fluctuation in unexpected affective shifts after accounting for lagged dynamics.

Population-Average NA–PA Dynamics

At the population level, participants reported higher typical PA than NA, consistent with work showing that average affective tone in community samples tends to be more positive than negative (Houben et al., 2015; Watson et al., 1988). Both NA and PA showed positive inertia, with AR(NA) slightly larger than AR(PA), suggesting that momentary elevations in NA persisted somewhat more strongly than elevations in PA at the hourly scale used here (Houben et al., 2015; Kuppens, Oravecz, & Tuerlinckx, 2010). Cross-lagged coupling was negative in both directions: higher NA predicted lower subsequent PA, and higher PA predicted lower subsequent NA. Interpreted as a descriptive bivariate regulatory system, these results are broadly consistent with reciprocal inhibition or spillover processes discussed in affect-dynamics theory (Kuppens, Allen, & Sheeber, 2010; Kuppens, Oravecz, & Tuerlinckx, 2010). At the same time, these coefficients should be interpreted as average short-term dependencies at a specific time scale (hourly, after detrending), not as mechanistic causal effects (Asparouhov & Muthén, 2018; Hamaker & Grasman, 2015).

Great job writing about affective dynamics in this section. In the paragraph above, can you provide some contrasts with previous results using similar models on the ADID data. For instance, I think You19a. However I think we only found significant fixed effect for CR from PA to NA there, but not from NA to PA (please check though). Note however that the discrepancy might stem from the use of a coarser averaging bin (4 times a day), as opposed to the hourly here. Also we did not have random effects, and thus missed some of the interesting findings here, such as covariances among inertia (BTW, make sure you define inertia earlier!) and CR coefficients.

Reviewers will probably ask you to add predictors of the interindividual differences and we do have some person-specific emotion regulation markers that could be used. But no need to do that now. Just keep that in mind for the revision/resubmission.

@articleYou19a, Author = You, Dongjun and Hunter, Michael and Chen, Meng and Chow, Sy-Miin, Doi = 10.1080/00273171.2019.1627659, Eprint = https://doi.org/10.1080/00273171.2019.1627659, Journal = Multivariate Behavioral Research, Note = PMID: 31264463, Number = 0, Pages = 1-25, Publisher = Routledge, Title = A Diagnostic Procedure for Detecting Outliers in Linear State-Space Models, Url = https://doi.org/10.1080/00273171.2019.1627659, Volume = 0, Year = 2019, Bdsk-Url-1 = https://doi.org/10.1080/00273171.2019.1627659

The population-average innovation covariance matrix further indicated that, after accounting for lagged dynamics, within-occasion unexpected shifts in NA and PA tended to move in opposite directions on average. The larger innovation variance for PA than for NA suggests that short-term unexplained fluctuations were greater for PA in this dataset.

Heterogeneity as a Substantive Finding, Not a Nuisance

The random-effects variance components indicated substantial between-person variability across the dynamic signature, including set points, lagged dynamics, and innovation parameters. The corresponding I^2 values were uniformly high, implying that the observed dispersion in Stage 1 estimates overwhelmingly reflected true between-person differences rather than sampling variability (Higgins & Thompson, 2002). Methodologically, this pattern motivates multivariate synthesis rather than edge-by-edge pooling, because heterogeneity was present across parameters and the off-diagonal elements of $\boldsymbol{\tau}^2$ showed how deviations in set points, inertia, coupling, and innovation structure co-occurred across individuals. Because $I_p^2 = \tau_p^2/(\tau_p^2 + \tilde{v}_p)$ in the present framework, values near 1 arise when true between-person heterogeneity is large relative to the typical Stage 1 sampling variance. In that sense, the uniformly high I^2 values here indicate that the spread in person-specific affect-dynamics estimates was driven primarily by genuine differences across individuals rather than by estimation noise.

Several examples illustrate this point. Individuals with higher NA set points tended to show greater NA inertia, suggesting that typical affective level and short-term persistence may be linked within persons. Likewise, stronger NA inertia was associated with more negative NA→PA coupling, and stronger PA inertia was associated with more negative PA→NA coupling, indicating that temporal persistence and cross-affect coupling were not independent sources of heterogeneity. Co-variation among the innovation parameters further suggests meaningful between-person differences in the magnitude and structure of within-occasion affective shocks.

Practical and Methodological Considerations

Several features of the analysis highlight common trade-offs in applied dynamic modeling. First, we imposed regular spacing by rounding responses to the nearest hour and inserting missing values for unobserved hours, and we applied within-person linear detrending to better align the data with a stationary VAR(1) approximation. These choices improve feasibility and interpretability within the present discrete-time framework, but they also mean that the empirical target is a detrended, approximately hourly affective process rather than the full irregularly spaced raw EMA record, shifting interpretation toward shorter-term fluctuations rather than slower drift across the month-long protocol (Guo et al., 2012; Li et al., 2022). Second, NA and PA composites were constructed from ordinal item ratings and treated as approximately continuous. Future work could examine whether latent measurement models or item-level analyses alter the estimated dynamic signature or its heterogeneity (Park et al., 2020; Watson et al., 1988).

Implications and Future Directions

Overall, the illustration supports the value of treating affect dynamics as person-specific signatures that can be meaningfully compared and synthesized (Fisher et al., 2017; Houben et al., 2015). Several extensions follow naturally from the same Stage 2 SEM template but are not pursued here, including covariate moderation of the parameter system and links between dynamic signatures and distal outcomes under full Stage 1 error propagation. Additional directions include allowing nonstationarity (e.g., regime switching or time-varying parameters; Chow & Zhang, 2013; Hutton & Chow, 2014), extending Stage 1 to continuous-time formulations to better handle irregular sampling without discretization (Park et al., 2025), and targeting multivariate constraints or derived functionals—such as stability summaries, impulse-response metrics, or innovation-based summaries—as higher-level quantities for synthesis.

General Discussion and Future Directions

The central aim of this paper was to develop a principled way to move from person-specific dynamic models to population-level inference without collapsing the distinction between within-person and between-person structure. MetaVAR addresses this aim by combining idiographic estimation in Stage 1 with multivariate meta-analytic structural equation modeling in Stage 2. The resulting framework preserves person-specific modeling while propagating first-stage uncertainty and supporting population-level inference on the full joint parameter system.

More broadly, the present results clarify where MetaVAR fits in the methodological landscape. MetaVAR is not intended as a universal replacement for one-step hierarchical modeling. Rather, it provides a principled two-stage alternative when researchers want to retain idiographic first-stage estimation while still drawing valid multivariate population-level inferences. In this sense, MetaVAR extends prior two-stage idiographic meta-analytic work to multivariate dynamic systems, where the scientific target is often not an isolated path but the joint parameter system. This perspective also aligns with the broader meta-analytic view that average effects should be interpreted alongside heterogeneity, not in place of it.

Several contributions follow from this framework. First, MetaVAR formalizes multilevel dynamic modeling as a meta-analytic problem in which persons play the role of studies. This makes it possible to synthesize not only average dynamics, but also between-person heterogeneity and cross-parameter co-variation across the dynamic system. Second, because Stage 2 is expressed as an SEM, the framework can accommodate model-level constraints, multivariate hypotheses, and extensions involving covariates or distal outcomes. Third, although the present manuscript focuses on observed-variable VAR(1) models, the same two-stage logic extends more broadly to Stage 1 models that yield a parameter vector together with an accompanying sampling covariance matrix.

The Monte Carlo results clarify where this framework is especially useful. MetaVAR did not uniformly outperform the comparison methods across all targets. For population-average parameters, the Bayesian multilevel VAR comparator was highly competitive and, in the present design, performed especially well in fixed-effect coverage. The clearest advantage of MetaVAR instead emerged for recovery of between-person heterogeneity and for the calibration of interval estimates for heterogeneity parameters. This distinction matters substantively because many questions in intensive longitudinal research concern not only what dynamic processes look like on average, but also how and why those processes differ across people.

The empirical illustration reinforces this point. In the ADID data, MetaVAR recovered interpretable population-average affect dynamics while also revealing substantial heterogeneity in set points, autoregressive persistence, cross-lagged coupling, and innovation structure. These results illustrate the value of treating a person's dynamic model as a multivariate signature that can be compared, related, and synthesized at the population level. More broadly, the illustration shows that heterogeneity is not merely a nuisance to be averaged away, but can itself be a target of explanation and theory building.

Move benchmarking results to Results section.

Our small-scale benchmarking results also highlight an important practical advantage of the MetaVAR workflow: computational scalability. Using a benchmark dataset with $N = 200$ individuals and $T = 200$ measurement occasions per individual, the naïve approach was the fastest method, MetaVAR occupied an intermediate position, and the joint Bayesian multilevel VAR approach was the slowest. Mean runtimes were approximately 14.72 seconds for naïve, 46.64 seconds for MetaVAR, and 363.42 seconds for BMLVAR. Thus, although MetaVAR incurs additional computation relative to a simple two-stage summary approach, it remained substantially faster than joint Bayesian estimation in this setting. This advantage is especially relevant for applications with large numbers of individuals because Stage 1 idiographic estimation is embarrassingly parallel: person-specific models can be fit independently across cores or computing nodes. As a result, the framework is well suited not only to larger N settings, but also to settings in which person-specific estimation becomes more costly, such as when each individual contributes longer time series or more complex dynamic models. Additional details are provided in the benchmarking vignette at https://jeksterslab.github.io/manMetaVAR/articles/benchmark.html.

At the same time, several limitations should be acknowledged. The simulation considered a bivariate data-generating process, fixed innovation covariance across persons, and correct model specification. The empirical illustration relied on observed composites, regularized hourly spacing, and a stationary VAR(1) approximation after detrending. Accordingly, the present results speak to performance under a favorable simulation setting and to one practically useful but simplified empirical workflow. Future work should examine higher-dimensional systems, heterogeneity in innovation processes, missing data and irregular spacing, model misspecification, and Stage 1 models beyond observed-variable VARs, including latent-variable, state-space, continuous-time, and nonlinear formulations. It will also be important to study the behavior of MetaVAR when Stage 1 sampling covariances are themselves approximate, such as when obtained from Taylor series approximations (e.g., delta method), sandwich estimators, or bootstrap procedures.

Cite some papers from the substantive literature that use distal outcomes.

Several extensions of the framework are especially promising. One is moderation: person-level covariates can be incorporated in Stage 2 to explain why some individuals exhibit stronger inertia, weaker regulation, or different innovation structures than others. Another is distal-outcome modeling, in which dynamic signatures are related to clinically or developmentally meaningful endpoints. A third is the synthesis of derived functionals rather than raw parameters, such as stability summaries, impulse-response quantities, network summaries, or other theory-relevant features of the dynamic system. These directions would extend the practical usefulness of MetaVAR as a synthesis framework for idiographic dynamical models.

A practical strength of the framework is that it is already implementable in accessible software. Stage 1 idiographic estimation can be carried out with the `fitVARMxID` package, which provides a convenient interface for fitting person-specific discrete-time and continuous-time VAR models in `OpenMx`; it is available on CRAN and includes support for multi-core computation to accelerate fitting across individuals. Stage 2 synthesis can be conducted with the `metaDyn` package, which implements MetaVAR and related extensions in an `OpenMx`-based workflow and is also available on CRAN. Together, these tools make the framework more readily usable in applied intensive longitudinal research.

Nice closing paragraph!!!

Overall, MetaVAR contributes a principled and flexible framework for drawing population-level conclusions from person-specific dynamics. Its most distinctive value is not that it replaces one-step hierarchical estimation in all settings, but that it provides valid multivariate population inference on heterogeneity and cross-parameter co-variation while preserving idiographic first-stage estimation. For intensive longitudinal research, this creates a practical route to asking not only what is typical, but also how people differ, how those differences cohere across a dynamic system, and how person-specific dynamic signatures relate to broader substantive outcomes.

Open Practices Statement

This study was not preregistered; however, to support transparency, reproducibility, and reuse, the analysis materials, software, and containerized workflows are publicly available on the Open Science Framework (https://osf.io/uenr7/) and GitHub (https://github.com/jeksterslab/manMetaVAR, https://jeksterslab.github.io/manMetaVAR/index.html). Source code and documentation for the `fitVARMxID` and `metaDyn` R packages are available on GitHub (https://github.com/jeksterslab/fitVARMxID, https://jeksterslab.github.io/fitVARMxID/index.html, https://github.com/jeksterslab/metaDyn, https://jeksterslab.github.io/metaDyn/index.html). Additionally, to facilitate reproducibility, we have provided containerized environments that include necessary dependencies to run the scripts. You can find detailed instructions on setting up and using these containers at: https://jeksterslab.github.io/manMetaVAR/articles/containers.html.

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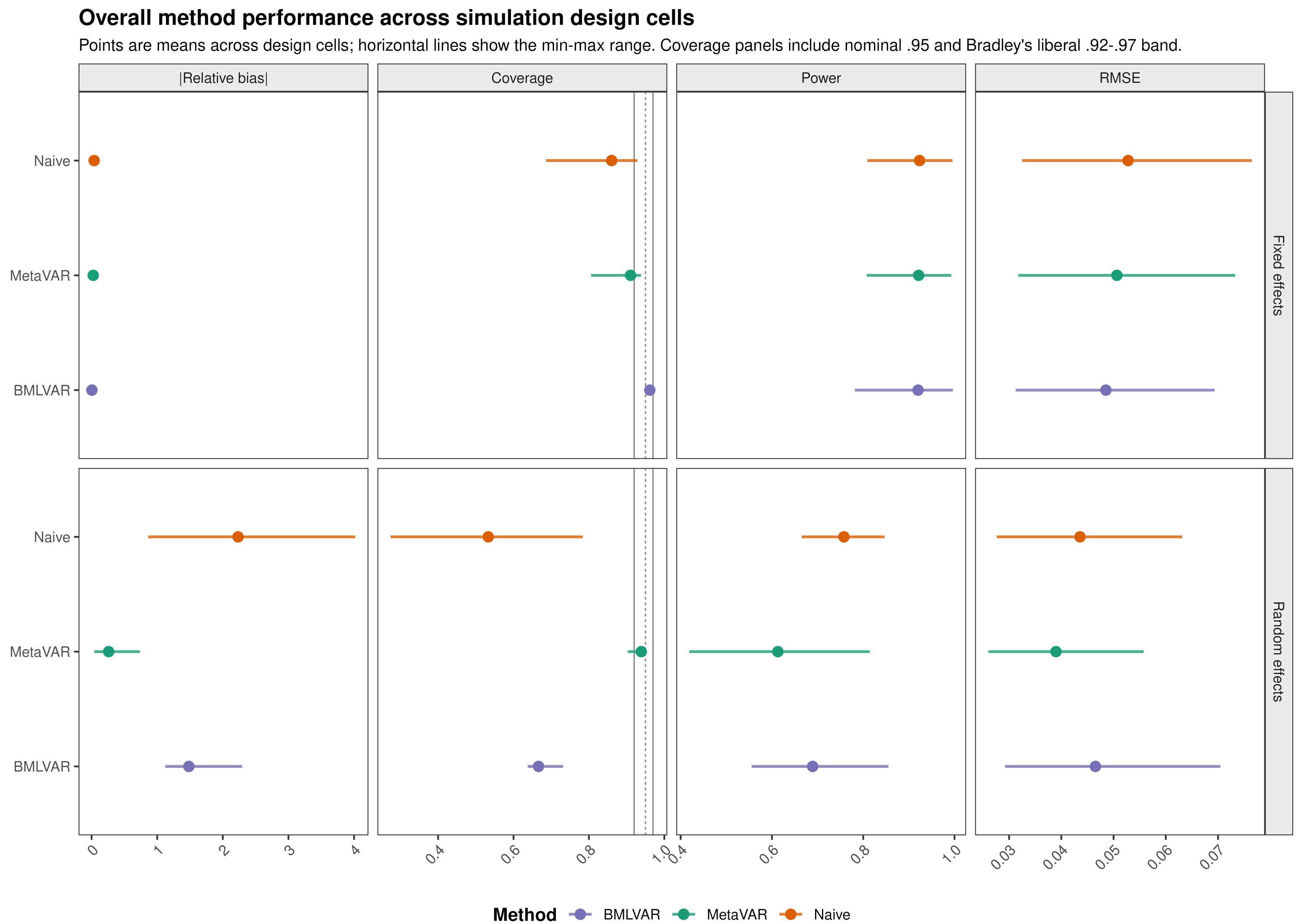
Figure 1

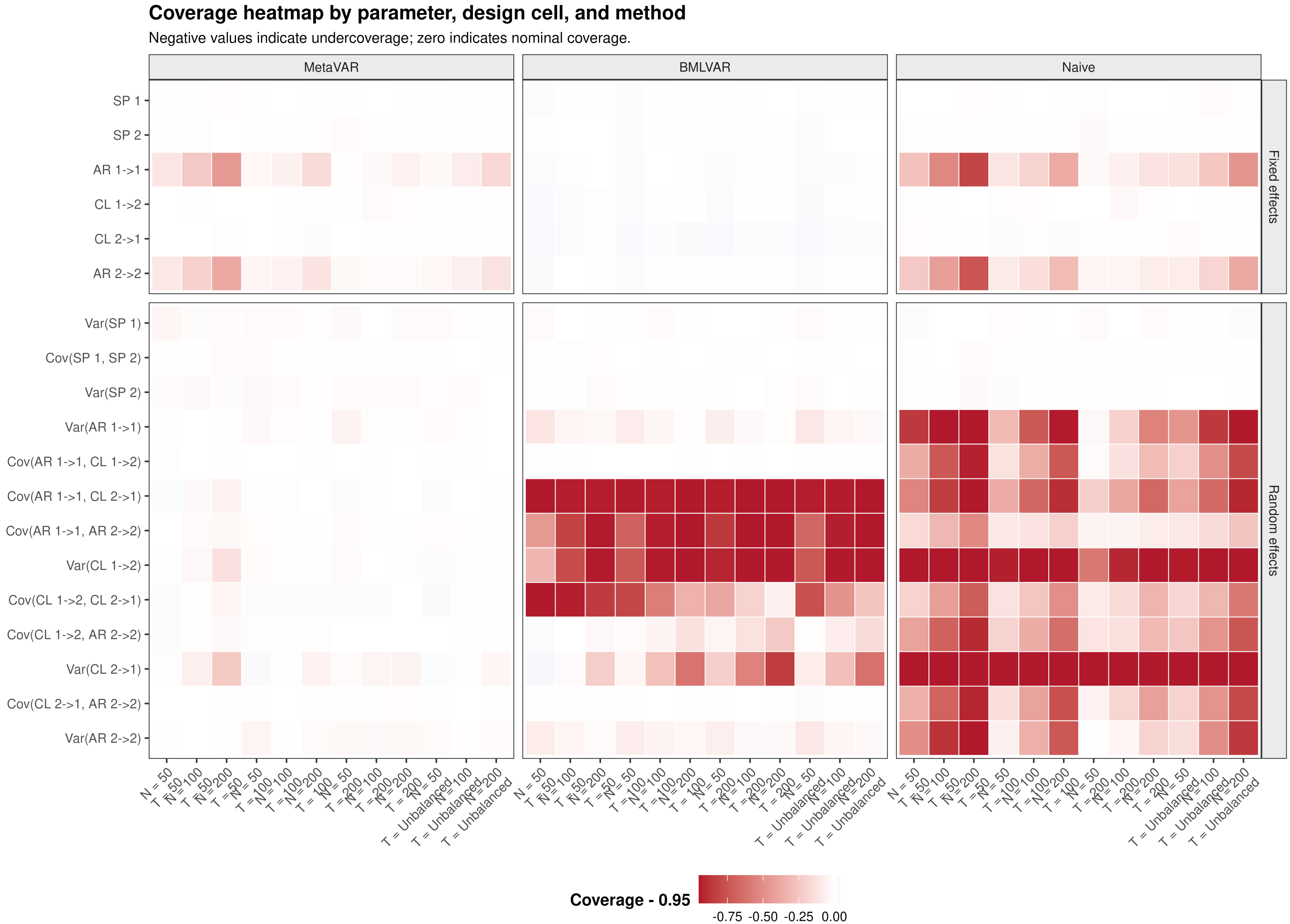
Overview of Monte Carlo Performance by Inferential Target and Method



Note: Each panel summarizes a performance metric across design cells separately for population-average parameters (fixed effects) and between-person heterogeneity parameters (random effects). Points denote the mean across design cells, and intervals denote the range across design cells. Across fixed effects, the three methods were broadly similar in point-estimate performance, with BMLVAR showing the best interval calibration. Across random effects, MetaVAR showed the clearest advantage, with lower absolute value of the relative bias and RMSE and coverage closer to the nominal 95% target than either BMLVAR or the naïve approach. For compact display in, parameter labels are abbreviated as follows: Sp = set point, AR = autoregressive effect, CL = cross-lagged effect, Var = variance, and Cov = covariance; subscripts follow the model notation, with 1 and 2 denoting the two variables.

Figure 2

Parameter-Level Empirical Coverage of Nominal 95% Intervals Across Methods and Design Conditions



Note: Cells show coverage relative to the nominal 0.95 benchmark for each target parameter under each simulation condition. Values near 0.95 indicate well-calibrated interval estimation, whereas lower values indicate undercoverage. The heatmap highlights two main patterns: BMLVAR performed best overall for fixed-effect coverage, whereas MetaVAR showed substantially better calibration for random-effects variances and covariances. Undercoverage was most pronounced for the naïve approach, especially for heterogeneity parameters and under shorter series. For compact display in, parameter labels are abbreviated as follows: Sp = set point, AR = autoregressive effect, CL = cross-lagged effect, Var = variance, and Cov = covariance; subscripts follow the model notation, with 1 and 2 denoting the two variables.